

Krishnaa Vadivel

Electrical Engineering | Computational Materials

New Haven, CT | srikrishnaa.careers@gmail.com | 516-853-5663

Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

LinkedIn: <https://www.linkedin.com/in/sri-krishnaa-ece-phd/>

Summary

Electrical-engineering PhD researcher combining semiconductor and photonic materials theory with numerical simulation, GPU/HPC software, and practical embedded-systems experience.

Technical Skills

Devices/materials: Semiconductor physics, many-body theory, exciton-phonon physics, integrated photonics, Meep FDTD, DOLFINx/PETSc

Compute: Python, C++, CUDA (cuBLAS, cuTENSOR), ROCm (rocBLAS, rocFFT), MPI, OpenMP, PETSc, SLEPc, Frontier, Perlmutter

Hardware/systems: Verilog, Lattice FPGA, PIC32, dsPIC, STM32, C8051, board debugging, ASP.NET, SQL Server

Experience

PhD Researcher

Yale University, Electrical Engineering

2021–present

- Develop first-principles and many-body models of self-trapped excitons and exciton-polarons in LiF and lead-free double perovskites.
- Implement distributed numerical workflows with MPI/OpenMP, PETSc/SLEPc, and CUDA/ROCm on Frontier and Perlmutter.
- Contribute semiconductor-material fabrication and characterization experience alongside simulation work in photonic and electronic materials.

Software and Embedded Systems Engineer

Probe Technologies / Weatherford Wireline Services, Houston, TX

2019–2021

- Integrated FPGA/MCU firmware, board debugging, CUDA/OpenCL diagnostics, and engineering-server software for downhole instrumentation.
- Used DOLFINx-based finite-element analysis for steel-pipe frequency investigations in a well-tool context.

Selected Projects

Integrated-photonics filter — Modeled a ring-resonator bandpass filter in Meep FDTD; the design was fabricated and tested with an erbium-doped fiber laser.

Semiconductor device modeling — Implemented Poisson, drift-diffusion, and pn-junction-style numerical models linking device theory to simulation.

Multiphysics finite elements — Built DOLFINx/PETSc heat, electrostatic, and incompressible-flow cases with visual post-processing.

Education

PhD, Electrical Engineering

Yale University

2021–present

MPhil and MS, Electrical Engineering

Yale University

2023

MEng and BS, Electrical and Computer Engineering

Cornell University,

2013–2018

Research Record

Co-author, *Journal of the American Chemical Society*, 2025; APS presentations, 2024–2026.